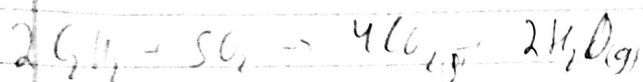
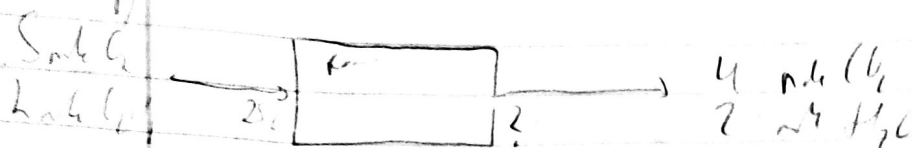


HW 11

$\dot{Q} = 0$ All $V_{i,k}$



Energy balance

$$\dot{H}_1 + \dot{E}_1^f - \dot{E}_2^f = \dot{Q} - \dot{W}$$

$$\dot{H} = 0 \quad \dot{H} = \dot{H}_f = \sum C_p \dot{T}$$

$$\sum n_i \hat{H}_i = \sum n_i \hat{H}_f$$

$$C_2H_6 \quad \Delta H_f^\circ = 228.2 \frac{kJ}{mol}$$

$$O_2 \quad \Delta H_f^\circ = 0 \frac{kJ}{mol}$$

$$CO_2 \quad \Delta H_f^\circ = -393.5 \frac{kJ}{mol}$$

$$H_2O \quad \Delta H_f^\circ = -241.83 \frac{kJ}{mol}$$

$$\dot{H}_1 (C_2H_6 \text{ entering}) = \int_{298.15}^{2413} C_{p,C_2H_6} dT = 0 \frac{kJ}{mol}$$

$$\dot{H}_2 (O_2 \text{ entering}) = 0 \frac{kJ}{mol}$$

1) 10

2) 20

3) 25

4) 10

$\frac{65}{100}$

$\frac{10}{25}$

$$\hat{H}_3 \text{ (kJ mol}^{-1}\text{)} = \int_{298.15}^{T_f} (a + bT + cT^2 + dT^3) dT$$

$$= \int_{298.15}^{T_f} (A + BT + CT^2 + DT^3) dT$$

$$= AT + \frac{BT^2}{2} + \frac{CT^3}{3} + \frac{DT^4}{4} \Big|_{298.15}^{T_f}$$

$$A = 14.10$$

$$B = 0.67344$$

$$C = -5.60 \cdot 10^{-5}$$

$$D = 1.715 \cdot 10^{-8}$$

$$AT_f + \frac{BT_f^2}{2} + \frac{CT_f^3}{3} + \frac{DT_f^4}{4} = 8706.61 \frac{\text{kJ}}{\text{mol}}$$

$$\hat{H}_4 \text{ (kJ mol}^{-1}\text{)} = \int_{298.15}^{T_f} (A + BT + CT^2 + DT^3) dT$$

$$A = 32.24$$

$$B = 0.001924$$

$$C = 1.055 \cdot 10^{-5}$$

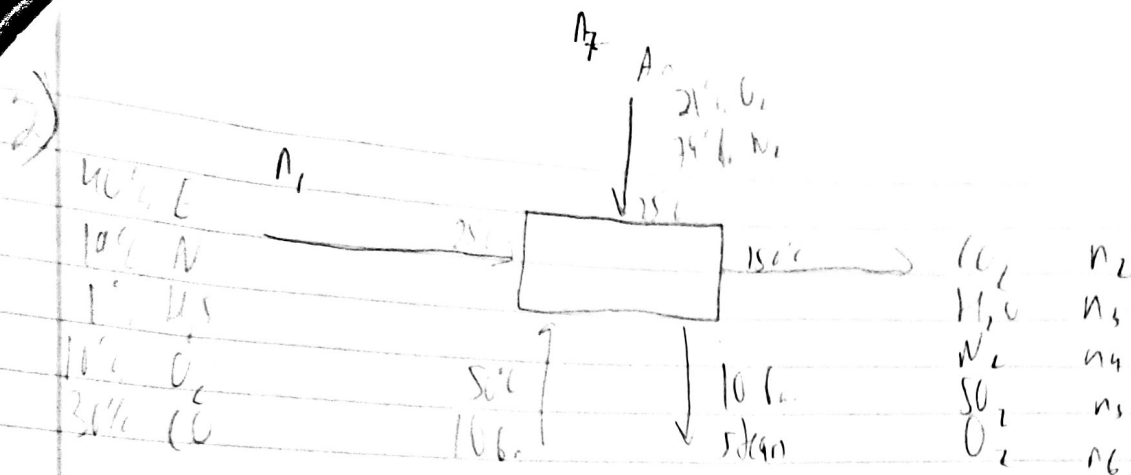
$$D = -2.596 \cdot 10^{-9}$$

$$= AT + \frac{BT^2}{2} + \frac{CT^3}{3} + \frac{DT^4}{4} \Big|_{298.15}^{T_f}$$

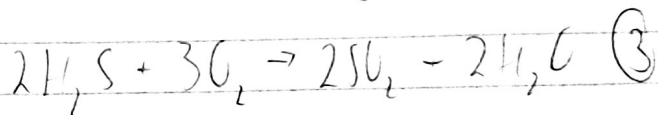
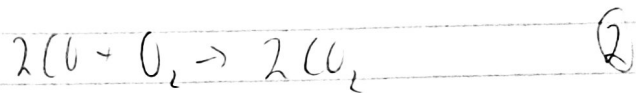
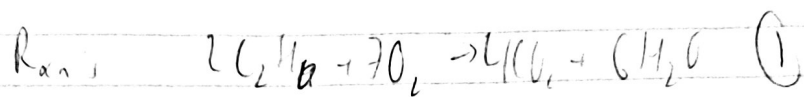
$$AT_f + \frac{BT_f^2}{2} + \frac{CT_f^3}{3} + \frac{DT_f^4}{4} = 9784.194006$$

$$2\hat{H}_3 + \hat{H}_4 = 2(228.2)$$

$$2\hat{H}_3 + \hat{H}_4 = 228.2$$



Assume: 100 mol basis for the whole



$$40 \text{ mol } C_2H_6 \rightarrow \frac{7 \text{ mol } O_2}{2 \text{ mol } C_2H_6} \rightarrow 140 \text{ mol } O_2$$

$$30 \text{ mol } CO \rightarrow \frac{1 \text{ mol } O_2}{2 \text{ mol } CO} \rightarrow 15 \text{ mol } O_2$$

$$1 \text{ mol } H_2S \rightarrow \frac{3 \text{ mol } O_2}{2 \text{ mol } H_2S} \rightarrow \frac{3}{2} \text{ mol } O_2$$

$$\text{Solid carbon } O_2 = 140 + 15 + \frac{3}{2} = 156.5 \text{ mol } O_2$$

$$1.2 \cdot 156.5 = 187.8 \text{ mol } O_2 \text{ in air}$$

$$n_{O_2} = 176 \frac{\text{gmol}}{\text{s}}$$

$$n_2 = \frac{187.8}{.21} = 894.29 \text{ mol air}$$

$$\checkmark n_{N_2} = 661.3 \frac{\text{gmol}}{\text{s}}$$

$$706.49 \text{ mol } N_2 \text{ in air}$$

$$\begin{aligned}
 L \quad \dot{n}_{L,0}^0 &= \dot{n}_{L,1} - 2\xi_1 & 40 &= 2\xi_1 & 20 \text{ m/s} &= \xi_1 = 40 \\
 (C) \quad \dot{n}_{C,0}^0 &= \dot{n}_{C,2} - 2\xi_2 & 30 &= 2\xi_2 & 15 \text{ m/s} &= \xi_2 = 30 \\
 (H) \quad \dot{n}_{H,0}^0 &= \dot{n}_{H,3} - 2\xi_3 & 1 &= 2\xi_3 & \frac{1}{2} \text{ m/s} &= \xi_3 = 1
 \end{aligned}$$

$$O_L \quad \dot{n}_{O_L,0} = \dot{n}_{O_L,1} - 7\xi_1 - \xi_2 - 3\xi_3$$

$$\begin{aligned}
 \dot{n}_{O_L,0} &= 10 + 157.8 - 7(20) - 15 - \left(\frac{3}{2}\right) \\
 &= 41.3 \text{ m/s } O_L = \dot{n}_L
 \end{aligned}$$

$$M_2 \quad \dot{n}_{M_2,0} = \dot{n}_{M_2,1}$$

$$\dot{n}_{M_2,0} = 19 + 706.49 - 725.49 \text{ m/s } M_2 = \dot{n}_4$$

$$H_1 \quad \dot{n}_{H_1,0} = \dot{n}_{H_1,2} + 6\xi_1 + 0\xi_2 + 2\xi_3$$

$$\dot{n}_{H_1,0} = 121 \text{ m/s } H_1 = \dot{n}_3$$

$$(C) \quad \dot{n}_{C,0} = 0 + 4\xi_1 + 2\xi_2 + 0\xi_3$$

$$\dot{n}_{C,0} = 110 \text{ m/s } C_L = \dot{n}_2$$

$$S_L \quad \dot{n}_{S_L,0} = \dot{n}_{S_L,1} + 2\xi_1$$

$$\dot{n}_{S_L,0} = 1 \text{ m/s } S_L = \dot{n}_5$$

$$\Delta \hat{h}_{f,c}^0 = -1428.6 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{h}_{f,o}^0 = -283 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{h}_{f,H_2O}^0 = -518.7 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{h}_i = 0$$

$$H = \int_{298.15}^{473.5} C_p dT$$

$$\Delta H = \sum_i \nu_i \Delta \hat{h}_{f,i}^0 + \sum_i \nu_i \Delta \hat{h}_{f,i}^0 - \sum_i \nu_i \Delta \hat{h}_i^0 + \sum \nu_e \hat{h}_e - \sum \nu_i \hat{h}_i$$

$$= 20(-1428.6) + 15(-283) + \frac{1}{2}(-518.7) + \sum \nu_e \hat{h}_e - \sum \nu_i \hat{h}_i$$

$$= -33076.35 + \sum \nu_e \hat{h}_e - \sum \nu_i \hat{h}_i$$

$$= -33076.35 + (110)(497) + (121)(1170) + (725.49)_{(3.0)} - (1)(5.27) + (41.3)(320)$$

$$= -29,171.8225 \frac{\text{kJ}}{\text{s}}$$

$$F_{in} \hat{h}_{in} - F_{out} \hat{h}_{out} = +29,171.8225 \frac{\text{kJ}}{\text{s}}$$

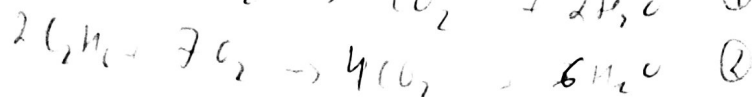
$$h(2722.1 - 210.9) = 29,171.8225$$

$$\frac{20}{25}$$

$$m = \frac{11.66}{110 \text{ mol gas}} \cdot \frac{1116 \text{ g}}{\text{mol gas}}$$

$$\alpha \quad m = 0.243$$

$$\frac{\text{kg steam}}{\text{kg gas burned!}}$$



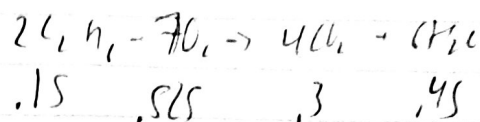
85% CH₄
15% C₂H₆

n₁

O₂ n₂ 25%



Assume a 1 mole basis for fuel.



Energy balance:

$$\Delta \hat{H} + \Delta \hat{E}_p - \Delta \hat{E}_k \approx 0$$

$$\Delta \hat{H}_{\text{com}} = -802.1 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{H}_{\text{c, (2)}} = -1428.1 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{H} = \Delta \hat{H}_{\text{com}} - \Delta \hat{H}_{\text{c, (2)}} = -682.21 - 214.24 = -896.5 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{H} = -896.5 - \Delta \hat{H}_1 - \Delta \hat{H}_2$$

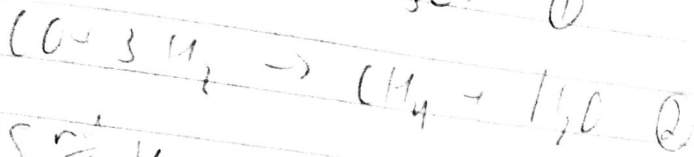
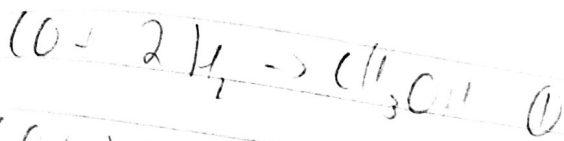
$$Q = -896.5 - (.075) \left(\frac{8.314}{1000} \right) (298.15)$$

$$= -896.7 \frac{\text{kJ}}{\text{mol}}$$

25
25

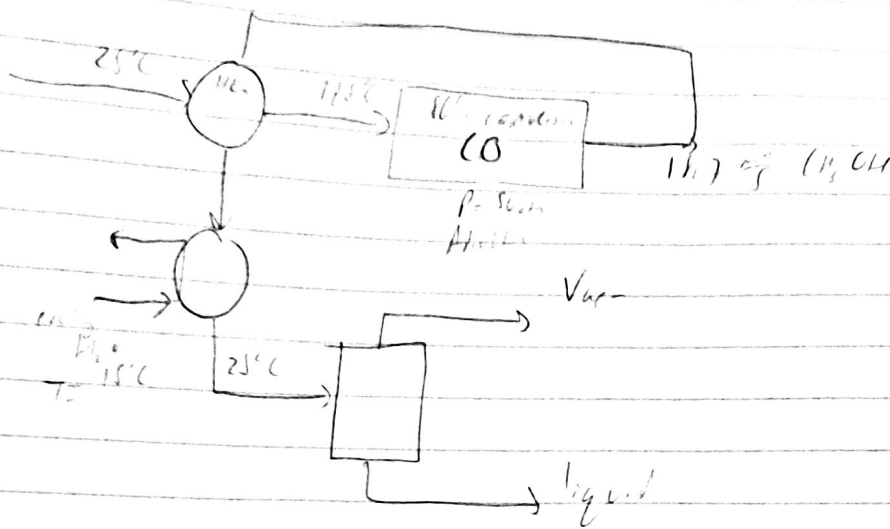
Bater

The results of Part a) & b) do not depend on the percent excess of oxygen because the reaction is being run isothermally. The excess air is not involved in the reaction.



$$75 \frac{\text{mol}}{\text{s}} H_2$$

$$25 \frac{\text{mol}}{\text{s}} CO$$



$$CO \quad \dot{n}_{CO,u} = \dot{n}_{CO,z} - \dot{\xi}_1 - \dot{\xi}_2$$

$$80 = \frac{\dot{\xi}_1 + \dot{\xi}_2}{25}$$

$$H_2 \quad \dot{n}_{H_2,u} = \dot{n}_{H_2,z} - 2\dot{\xi}_1 - 3\dot{\xi}_2$$

$$20 = \dot{\xi}_1 + \dot{\xi}_2$$

$$CH_4 \quad \dot{n}_{CH_4,u} = \dot{n}_{CH_4,z}^0 + \dot{\xi}_1$$

$$20 - 18.7 = \dot{\xi}_2$$

$$H_2O \quad \dot{n}_{H_2O,u} = \dot{n}_{H_2O,z}^0 + \dot{\xi}_2$$

$$1.3 = \dot{\xi}_2 \frac{\text{mol}}{\text{s}} \quad \checkmark$$

$$18.7 = \dot{\xi}_1 \frac{\text{mol}}{\text{s}}$$

$$\dot{n}_{H_2O} = 25 - 18.7 - 1.3 = 5 \frac{\text{mol}}{\text{s}} H_2O$$

$$\dot{n}_{H_2O,u} = 75 - 2(18.7) - 3(1.3) = 33.7 \frac{\text{mol}}{\text{s}} H_2$$

$$\dot{n}_{CH_3OH} = 18.7 \frac{\text{mol}}{\text{s}} CH_3OH$$

$$\dot{n}_{H_2} = 1.3 \frac{\text{mol}}{\text{s}} H_2 \quad \checkmark$$

$$C_{1H_4} C_p: 19.25 + .08213T + 1.147 \cdot 10^{-5} T^2 + -1.137 \cdot 10^{-8} T^3$$

$$C_{4CH} C_p: 21.15 + .07092T + 2.087 \cdot 10^{-5} T^2 + -2.852 \cdot 10^{-8} T^3$$

$$H_2O C_p: 32.24 + .01927T + 1.055 \cdot 10^{-5} T^2 + -3.596 \cdot 10^{-9} T^3$$

$$H_2 C_p: 27.14 + 0.0693T + -1.381 \cdot 10^{-5} T^2 + 7.645 \cdot 10^{-9} T^3$$

$$CO C_p: 30.87 + -.01285T + 2.789 \cdot 10^{-5} T^2 + -1.272 \cdot 10^{-8} T^3$$

$$\Delta H = \sum n_c \bar{h}_c - \sum n_i \bar{h}_i$$

$$\frac{10}{25}$$